
AR 508: Marine Robotics

Course Code	AR 508
Course Name	Marine Robotics
L-T-P-C	3-0-0-3
Prerequisites	Consent of faculty advisor
Intended for	UG, PG and PhD
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Types of Marine robots and applications:** Introduction to the types of marine robots, Classification based on applications, Approach for deployment, operation and maintenance of marine robots such as surface vehicles and underwater vehicles (ASVs, AUVs, ROVs, underwater gliders and floats). (4 hours)
- **Vehicle design:** Mechanical design of marine robots, structures, materials, pressure hull, Vehicle parameters and performance metrics including pressure, buoyancy, stability, ballasting, propulsion, power, speed, range, and cost of transport (COT). (8 hours)
- **Vehicle mathematical modelling:** Classification of models, Rigid body Kinematics, frame transformations between body, flow, and non-accelerating frames, Euler angles, quaternions, Rigid-Body Kinetics, Equations of motion (Linear and nonlinear), Hydrostatics, maneuvering model, coupled motion model, environmental disturbances. (8 hours) Navigation: Marine Sensors and navigational strategies for localization using dead-reckoning, SLAM and uncertainty/probabilistic approaches, and Observer-based design. (6 hours)
- **Guidance:** Path planning algorithms and path following strategies include line of sight guidance strategies, pure pursuit guidance, constant bearing guidance, and trajectory tracking. (6 hours)
- **Control:** Modelling and control using PID controllers, open-loop stability, and state feedback control, maneuverability system architectures, and actuator models. (6 hours)
- **Final project:** Student project towards modelling of a marine robot using MATLAB. (4 hours)

Text Books

1. **Handbook of Marine Craft Hydrodynamics and Motion Control**, 2nd Edition, Wiley.
2. Beard, R. W. and T. W. McLain, **Small Unmanned Aircraft: Theory and Practice**, Princeton University Press.
3. Moore S.W., Bohm H., and Jensen V., **Underwater Robotics: Science, Design, and Fabrication**.

References

1. Triantafyllou MS, Franz S. Hover, **Maneuvering and control of marine vehicles**, Lecture Notes, Department of Ocean Engineering Massachusetts Institute of Technology Cambridge, Massachusetts USA.
2. Antonelli, G., **Underwater robots In Encyclopedia of Systems and Control**, (pp. 2384-2388), Cham: Springer International Publishing.